Sankalp Jha

Software Engineer - C, Networking & Low-Level Systems
Delhi NCR, India

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Education

Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	**7.25/10**
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	**91.2%**

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (***May 2025 – Sept 2025***)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4 *P4, C++, Python, P4Runtime, Docker, Intel Tofino*

- Architected a switch-native framework compiling ML decision trees directly into P4 data planes using Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs, working around hardware ASIC constraints on Intel Tofino silicon.
- Built a C++/P4Runtime data-path testing pipeline spanning Tofino hardware and Mininet software targets, improving deployment reliability by **30%**.
- Optimized low-level SID recirculation logic to hit **10** Gbps line-rate throughput with up to **5x** lower inference latency than control-plane inference.

Open Source Contributor - WasmEdge/WasmEdge (C/C++) (*2024 – Present*)

- Diagnosed and fixed a WASI clock bug in the C++ runtime causing **100%** CPU usage on macOS using gdb and perf; merged to main runtime. (PR **#4470**)
- Built a standalone Alpine-base Docker image build pipeline for the runtime. (PR **#4466**)
- Upgraded the base Alpine image to **3.23** and restored static LLD linker support. (PR **#4440**)

- facebook/bpfilter ()

- build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c (PR **#507**)

Open Source Contributor - Project-HAMi/HAMi (CNCF Kubernetes GPU Device-Sharing) (*2024 – Present*)

- Fixed scheduler and device-plugin bugs in HAMi, a CNCF project for GPU/device resource sharing and isolation across Kubernetes clusters - **5** PRs merged.

Projects

XNIC v1 | *C, Linux Kernel, PCI, DMA, NAPI, QEMU, DPDK*

- Built a clean-room Intel 82540EM/e1000-compatible PCI Ethernet driver in C: PCI probe/remove, BAR0 MMIO, coherent and streaming DMA rings, interrupt-driven NAPI, TX backpressure, ethtool telemetry, and watchdog/reset recovery.
- Qualified it under KFENCE and sparse with staged fault injection: **646,400** packets at zero loss, **1,000** interface cycles, **100** rebinds, and **100** reset-under-traffic cycles on reproducible QEMU/HVF automation.
- Added a W5500 SPI driver path and a gated DPDK L2 forwarder; validated partial-TX cleanup and signal-safe shutdown against the net_pcap virtual PMD.

ebpf-mcp-tracer | *Python, bpftrace, eBPF, Linux*

- Secure server exposing controlled Linux kernel tracing to LLM agents: strict probe allowlist, dry-run validation, and hard timeouts so agents can debug system bottlenecks without raw unsafe eBPF access.

Los-Tecnicos | *Go, TypeScript, Rust, Soroban, Raspberry Pi, ESP32*

- Decentralized energy-grid system with a Raspberry Pi/ESP32 hardware mesh; built the Go backend and **4** Soroban smart contracts for SW/HW bring-up; won Asia-Pacific Web3 bounties, recognized by the Government of India.

Technical Skills

- **Languages:** C, C++, Python, Go, P4, JavaScript, TypeScript, Rust
- **Systems / Networking:** Linux (x86_64, ARM64, AMD64), eBPF, gRPC, Protobufs, Scapy, socket-level networking
- **Debugging / Profiling:** gdb, perf, pprof, htop
- **Tools / Platforms:** Docker, CI/CD, GitHub, GitLab, UTM
- **Databases:** SQL, PostgreSQL, MySQL, MongoDB, Redis, GraphQL

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," ***2026** ICCSC*, Ghaziabad, India, Feb. **2026**. DOI: 10.1109/ICCSC67078.2026.11468597

Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (***May 2024 – Dec 2024***)
  – Led **6+** writers and **4** designers on documentation and event content; social media revamp lifted engagement **60%**.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (***Nov 2023 – Nov 2025***)
  – Co-led **10+** tech events; anchored GDSC WOW at IIIT Delhi (**800+** live audience, **31+** crew).

Achievements & Certifications

- **2x** AKTU State TT Gold (**2023**, **2024**); Robotics Olympiad Bronze, Thailand (**2019**); NASA, NPTEL & ISRO certifications